SASANGA ATHURALIYA

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PROFILE

I am an Automotive Engineering graduate (B.Sc. Hons) specializing in chassis design, topology optimisation, and aerodynamics. I have hands-on experience in vehicle diagnostics and maintenance from industrial training at Diesel & Motor Engineering PLC, and advanced CAD proficiency in SOLIDWORKS, CATIA and Fusion. My final-year research focused on biomorphic generative design for Formula Student chassis optimization. I combine technical engineering expertise with leadership in cross-functional teams, demonstrated through leading the Society of Automotive Engineering and the PhoneSim engineering team. I aspire to contribute to innovative automotive and motorsport engineering solutions.

TECHNICAL SKILLS

- Computer-Aided Design
- Finite Element Analysis
- Computational fluid dynamics
- Topology Optimisation
- Vehicle Modelling & Simulation
- Vehicle Design & Development
- Vehicle Testing & Validation
- Vehicle Performance Analysis
- Vehicle Scanning & Diagnosing
- Vehicle Repairing & Maintenance

SOFT SKILLS

- Project Management
- Team Leadership
- Technical Communication
- Problem Solving
- Time Management
- Adaptability
- Creativity

LANGUAGE

- Sinhala
- English

VOLUNTEER

- **General Member**
  Apr 2024 - Present
  Ceylon Motor Sports Club (CMSC), Colombo
- **General Member**
  Oct 2022 - May 2026
  Rotaract Club of CINEC Campus, Malabe

EDUCATION

- **G.C.E. Ordinary Level Examination (2018)** — **Jan 2008 - Dec 2018**
  Ananda College, Kottawa
- **G.C.E. Advanced Level Examination (2021)** — **Jul 2019 - Feb 2022**
  Royal College, Colombo 07
- **B.Sc. (Hons) in Automotive Engineering** — **Oct 2022 - Oct 2026**
  CINEC Campus, Malabe
  Soon to be Graduated (~292 ECTS)

WORK EXPERIENCE

- **Engineering Undergraduate Representative** — **Oct 2022 - Oct 2026**
  CINEC Campus, Malabe
  Act as liaison between students and faculty, addressing academic concerns, organising batch events and ensuring a positive learning environment.
- **Assistant Club Service Director** — **Jun 2023 - May 2024**
  Rotaract Club of CINEC Campus, Malabe
  Led service initiatives and supported club operations, helping the club win recognition and awards during the year. Focused on fellowship, engagement, and impact.
- **President** — **Mar 2025 - June 2026**
  Society of Automotive Engineering of CINEC Campus, Malabe
  Co-founded and led the society to unite automotive undergraduates. Oversee events, industry connections and project development to promote automotive technical excellence.
- **Automotive Engineering Trainee - Internship** — **Jul 2025 - Jan 2026**
  Diesel & Motor Engineering PLC, Colombo
  Gaining hands-on experience in vehicle diagnostics, engine systems and maintenance procedures at a leading automotive service provider.

CERTIFICATIONS

- **Workshop Practice 1 - WP1** — **Nov 2023**
  Ceylon German Technical Training Institute, Moratuwa
- **SOLIDWORKS CAD Design Associate - CSWA** — **May 2025**
  SOLIDWORKS, Dassault Systèmes
- **SOLIDWORKS CAD Design Professional- CSWP** — **Jul 2025**
  SOLIDWORKS, Dassault Systèmes

ACADEMIC PROJECTS

- **Final Year Individual Research Project** **Jan 2026 - August 2026**

  Biomorphic Generative Design and Lattice-Integrated Optimisation of a Formula Student Chassis for Enhanced Performance.
  - Objective: Enhance stiffness-to-weight ratio through biomorphic topology optimisation and lattice integration while ensuring FS-UK 2025 regulatory compliance.
  - Completed baseline steel spaceframe CAD in 3DX CATIA with full FS rule validation (zero violations). Conducted sectioned topology optimisation (front, mid, rear sections) with material comparison (AISI 4130 vs Ti-6Al-4V LPBF Post-HIP) in 3DX SIMULIA. Established comprehensive FEA validation across 7 load cases (torsional stiffness, bending, von Mises stress, impact, buckling).
  - Prototype: 1:3 scale 3D-printed geometric validation model confirming design feasibility. Methodology validated and confirmed & partial topology optimisation results achieved despite complex computational and feature-conflict limitations.
  - Target outcomes: 30% mass reduction, 2.0+ safety factor, 15% stiffness-to-weight improvement.
  - Status: Core methodology validated & framework established for future continuation as the project will be independently developed post-graduation.
- **Final Year Group Research Project** **Feb 2025 - Mar 2025**

  PhoneSim: Smartphone-Integrated Driving Simulation Controller System

  As Engineering Director led mechanical design, system architecture, and validation. Developed integrated system leveraging smartphone gyroscope, vibration motor, and display combined with custom steering wheel and 3-pedal assembly. Engineered ESP32-S3 controller bridge integrating smartphone IMU input with Windows/vJoy game controller protocol. Completed three-generation design progression (Gen 1.1 Concept, Gen 1.2 Refinement, Gen 1.3 Design-for-Manufacture) with comprehensive CAD development to deliver functional prototype testing with pedal sensing, steering input calibration, and game integration validation. Multi-disciplinary collaboration across 5-person team coordinating mechanical design, manufacturing, and system integration.
- **F1 Model for Aerodynamic Performance Analysis Using CFD** **Feb 2025 - Mar 2025**

  Conducted a CFD-based aerodynamic study of three Formula 1 car models designed in SOLIDWORKS, focusing on drag coefficient (Cd) optimisation for high-speed performance. Simulations were performed under controlled conditions.
  - Model 01 (Base): Cd = 0.826
  - Model 02 (with DRS): Cd = 0.791
  - Model 03 (with DRS + Splitter): Cd = 0.773

  Despite limitations, the results reflected realistic aerodynamic trade-offs observed in F1, where Cd values between 0.7–1.2 are typical to balance drag and downforce. The project highlights the role of design modifications like DRS and aerodynamic splitters in optimizing performance and stability in professional motorsport applications.
- **Shock Spoke Wheel Mechanism for Enhanced Motorbike Suspension** **May 2025 - Jun 2025**

  Designed and analyzed a novel wheel with integrated shock-absorbing spokes to enhance ride comfort on rough roads. Applied SOLIDWORKS for 3D modeling and FEA for structural behavior under dynamic loads.

NON-ACADEMIC PROJECTS

- **Project Oceanus 2.0 & 3.0** **Apr 2023 - June 2023**

  Rotaract Club of CINEC Campus

  Multi-phase marine and coastal ecosystem protection initiative in collaboration with National Cleaner Production Centre (NCPC) and Ocean University, Tangalle. Phase I: Expert-led awareness session on coastal ecosystems and marine biodiversity. Phase II: Beach cleanups (200+ kg plastic collected, custom sand-sifting sieves developed and donated to NCPC). Phase III: Mangrove restoration and sea turtle conservation (300+ eggs secured and protected). Featured in Rotary Projects Around the Globe (November 2023). Demonstrates hands-on environmental stewardship and community collaboration at scale.

  Recognition: Featured in Rotary Projects Around the Globe – November 2023 Edition by Rotary International
- **Project Numinous** **July 2023 - Jun 2024**

  Rotaract Club of CINEC Campus

  Led a community initiative with four phases to celebrate and showcase the equal rights and cultural diversity of major religions in Sri Lanka, including Buddhism, Hinduism, Islam, and Christianity. The project promoted mutual respect, peace, and social harmony through events and awareness campaigns involving students and local communities, enhancing intercultural understanding and unity on campus.

  Bronze Award: Most Outstanding Club Service Initiative Fellowship (Internal) at the 34th Rotaract District Assembly 2023/24
  Silver Award: Most Outstanding Club Service Initiative Fellowship (Internal) at the 35th Rotaract District Assembly 2024/25
- **Project Randoli** **Jun 2024 - Aug 2024**

  Rotaract Club of CINEC Campus

  Chaired the first phase of Project Numinous focusing on Buddhist culture during Kandy Esala Perahera. Provided hands-on cultural volunteering experience, strengthened internal fellowship, and promoted awareness of Sri Lanka's religious heritage as foundation for broader interfaith activities.

  Gold Award: Most Outstanding Club Service Initiative Fellowship (Internal) at the 35th Rotaract District Assembly 2024/25

REFERENCES

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